

Unit 1 :Population and Sample

*** Statistical population (population) –**

A statistical population can be defined as the collection of units (individuals) satisfying particular property.

OR A group of individuals under study is known as population

Population may consists of finite or infinite number of units, accordingly it is called as finite or infinite population. e.g. students in the class, employee in industry, insects in a field, persons suffering from tuberculosis in city etc. are the examples of finite population, all real numbers, stars in the sky, trees in jungle etc. are the examples of infinite population.

Generally, the population consist of large number of animates and inanimates. Moreover the units constituting the population may vary from survey to survey depending upon the aims and objectives of the survey. In brief one should keep in mind that population is not only human population but it is generally a group or a collection of units specified certain characteristics.

* Census method :-

The procedure of collecting the data from entire population i.e. from every member of the population is called Census method or complete enumeration or 100% inspection.

Limitations of Census method:-

1. if population is infinite Census method is not applicable. E.g. it is not possible to test the quality of every grain of wheat in a bag.
2. there are some situations where Census is possible but impracticable. E.g. testing of blood, to test entire blood of an individual is possible but not necessary.
3. when items are destroyed under investigation, in that situation Census is not applicable. E.g. testing of crackers and explosives etc.
4. Census method provides reliable results but due to voluminous work it is not applicable when results are required in short time as well as resources for survey like trained persons and money are limited.

*** Sampling method:-**

To overcome the limitations of Census method, sampling is used. Sample is a part or sub-group of population selected on some basis. Sample consist of few items of a populations such that it is true representative of the population.

Sample – A part of population under study is called Sample

OR a finite sub-group of population is called sample.

Sampling – sampling is a procedure by which a representative sample is selected from population on some basis for required statistical investigation.

Illustration

1. while purchasing food grains, we inspect only a handful of grains and draw the conclusion about quality of whole lot. In this case handful of grains is a sample and whole lot is the population.
2. while examining blood of an individual, few drops are taken out for diagnosis. In this case drops taken for diagnosis is a sample and entire blood in the body is the population.

Advantages of sampling over Census method

- 1) Sampling is less time consuming – as compare to population. sample consist of less no of units, so time required for collection and analysis of data is reduced and results are obtained quickly.
- 2) Sampling is economical – since sample is only a part of population, therefore expenses over collection and analysis is always less than those in Census.
- 3) Sampling increases accuracy – as compare to Census method only limited number of units are to be processed in a sampling method, so sample can be studied carefully and detailed information is collected also we can use sophisticated machinery, well trained staff hence accuracy of result can be increased.
- 4) Sampling has greater scope – if population is infinite as well as if units are destroyed in the analysis Census method is not applicable so sampling has greater scope in this situation.

* Sampling methods:-

- 1) simple random sampling
- 2) stratified sampling
- 3) systematic sampling

1) Simple random sampling (SRS)-

SRS is a procedure of drawing a sample from population, in which every unit in the population has equal chances (probability) of being selected in the sample.

This is simple and most commonly used method. There are two type of SRS due to slite differences in the procedures of selecting the units.

a) Simple random sampling with replacement (SRSWR)-

In this method units are selected one by one such that a unit selected at a time is replaced back to the population before drawing the next unit. This procedure is continued until a sample of required size is get selected.

Due to replacement it may happen that same unit may be included more than ones in the sample, so if the required sample size is 'n', the effective sample size some times less than 'n'. In a sampling from a population of size 'N', the probability of selecting a unit at each draw is $(1/N)$. This method is rarely used.

b) Simple random sampling without replacement (SRSWOR)-

In this method units are selected one by one such that a unit selected ones is not replaced back to the population before drawing the next unit. This procedure is continued until a sample of required size is get selected.

Here same unit will not be included more than ones in the sampling. In a sampling from a population of size 'N', the probability of selecting a unit at r^{th} draw is $(1/(N+1-r))$. SRSWOR is more suitable method for drawing a representative sample when population is more or less homogeneous.

Illustration

- 1) To test quality of food grains in the bag, we take a handful of grains for testing is a SRSWR
- 2) Testing human blood by taking few drops from body is SRSWOR
- 3) In order to find average life of a bulb, one can use SRSWOR

2) Stratified sampling :-

When population is not homogeneous, SRS method does not provide proper representative sample. In that situation the total population is divided into different groups according to some criteria, so that the units within the group are homogeneous and units between any two groups are heterogencous. These groups are known as strata. The samples of suitable sizes

are selected from each stratum according to relative importance of the stratum in population by SRS method. The required sample is the aggregate of the sampled units from each of the stratum. The procedure of drawing such a sample is known as stratified sampling.

Note- stratified sampling is more representative because it gives proper representation to every group of population

Illustration-

- 1) In order to estimate crop yield we can divide the field under cultivation in plots according to fertility of soil, here each plot is considered as a strata.
- 2) Suppose proportion of defective items is to be estimated in a manufacturing process. In this case we can use stratified sampling by taking strata as production in different shifts.

3) Systematic sampling:-

When the population units are occur in a sequence and a sample of size 'n' is to be draw, the total population is divided into 'n' sequential groups. That is population size is 'N' and the units in the population are arranged sequentially 1,2,3.....N. Then to draw a sample of size 'n' divide the population in 'n' sequential groups each having $N/n = k$ (say) units. Thus 1st group contains units having serial number 1 to k, 2nd group contains units having serial number (k+1) to 2k, 3rd group contains units having serial number (2k+1) to 3k,and so on. Now to draw a sample, select a unit by SRS method from the 1st group, suppose ith unit is selected, then take (i+k), (i+2k), (i+3k), (i+(n-1)k) serial numbered unit from the subsequent group, it forms a sample of size 'n'. Such a selected sample is known as systematic sampling.

Note that when units in population were occurs cyclically and if 'k' is equal to value of period of cycle then this method does not provide representative sample.

Illustration -

- 1) To select house for statistical survey, we can use house No's. In this case systematic sampling is used.
- 2) To study the purchasing habit of people, we can use bill No's from bill book. In this case systematic sampling is used.

Assignment No. - 1

- Q1. Define - i) population and sample with illustration.
- Q2. Define census method and explain its limitations.
- Q3. Define sample, sampling and explain advantages of sampling over census method.
- Q4. Explain simple random sampling method and give one illustrated example.
- Q5. Explain stratified sampling method and give one illustrated example.
- Q6. Explain systematic sampling method and give one illustrated example.

Q.7. Choose most correct alternative.

1. If population is studied by collecting the information from each and every member of population is called.....
 - a) Census method b) sampling method c) stratified sampling method d) none of these
2. If population is infinitely large, then it is studied by method only.
 - a) Sampling b) census c) SRSWOR d) SRSWR
3. If units in the population are destroyed under investigation, then it is studied by method.
 - a) SRSWOR b) SRSWR c) stratified sampling d) all of these
4. Few drops of blood taken for diagnosis is an example of
 - a) SRSWOR b) SRSWR c) SRS d) none of these

5. Let A: sampling is time consuming and B: sampling is economical. then
 a) A & B both are true b) only A is true $\neg$ c) only B is true d) none of these
6. In method of sampling every unit in the population has equal chances of being selected in the sample.
 a) SRS b) stratified c) systematic d) all of these
7. If population is heterogeneous, then Sampling method provides most representative sample.
 a) SRS b) stratified c) systematic d) all of these
8. For application of systematic sampling method the population.....
 a) Units must occur in sequence b) is homogeneous c) is heterogeneous d) none of these
9. Units between any two stratum are
 a) Homogeneous b) heterogeneous c) equal in number d) all of these
10. In Method of sampling same unit may be selected more than once.
 a) SRSWOR b) SRSWR c) SRS d) all of these.
11. SRS method provides a representative sample if population is -----
 a) homogeneous b) heterogeneous c) (a) and (b) both d) none of these
12. Let A: units within a strata are homogeneous,
 B: units between any two strata are heterogeneous.
 a) both A and B are true b) only A is true c) only B is true d) none of these.